

Working with Behaviour Trees

To implement a Behaviour Tree in UE5, you would usually start by defining a Blackboard. This is a data storage system that the AI character uses to make decisions. It might include data like the location of the player, the AI's current health, or whether the AI has seen an enemy. Next, you can create the Behaviour Tree itself in the UE5 editor, using the various components to define your AI's possible actions and decision-making process. Tasks and Decorators can be created using Blueprint Visual Scripting, allowing you to define complex behaviors without writing code.

Navigation System

In video games, navigating the world is a critical aspect of AI behavior. It allows non-player characters (NPCs) to move around the environment in a way that seems intelligent and reactive to the game world's conditions. To facilitate this, Unreal Engine 5 (UE5) comes equipped with a powerful Navigation System.

Understanding Navigation Meshes

At the heart of the Navigation System is the concept of a Navigation Mesh (NavMesh). The NavMesh is a data structure that represents the "walkable" areas of a game world. It's a mesh that covers the level, defining where AI characters can move. The system generates this by analyzing the environment and creating polygons (NavMesh Bounds Volumes) over accessible surfaces.

Generating a NavMesh

To use the Navigation System, the first step is to generate a NavMesh for your level. You do this by placing a "NavMesh Bounds Volume" in your level, which defines the area where the NavMesh will be generated. The engine will then automatically create the NavMesh within this volume, covering all walkable surfaces.

Navigation Agents and Their Properties

Navigation Agents are the entities that use the NavMesh to navigate. They can be NPCs, players, or anything else that needs to move around. In UE5, you can set properties for these agents, like their radius and height, which affects how they navigate the world. For instance, a small agent might be able to move through narrow spaces that a larger agent can't.

Navigation Queries and Pathfinding

The Navigation System in UE5 uses pathfinding algorithms to find the shortest path for an agent from its current location to a destination, using the NavMesh. It's capable of complex pathfinding tasks, including dynamic obstacles avoidance. For this purpose, UE5 uses a modified A* algorithm for pathfinding and navigation queries.

Dynamic Navigation

UE5 also supports dynamic NavMeshes. This means that if

the game environment changes - for instance, a door opens or a bridge collapses - the NavMesh can update in real-time to reflect these changes, allowing AI characters to react to the changing environment.

Basic Components

Unreal Engine 5 (UE5), like its predecessors, is based on a component-driven architecture. This means that everything you see in the game world - from characters to lights to scenery - is created using various components. Let's explore some of these fundamental components in UE5.

Components

Actors are made up of Components. For example, a player character Actor might consist of a static mesh component for the character's body, a camera component for the player's viewpoint, and a movement component to control the character's motion.

Static Mesh Components

Static Mesh Components are one of the most commonly used components in UE5. They represent a 3D model (a mesh) with an associated material (texture). The model and material are usually created in a 3D modeling program like Blender or Maya, then imported into UE5.

Light Components

Light Components are used to create and control lighting in a UE5 game world. There are several types of light components available in UE5, including point lights (which emit light in all directions from a single point), directional lights (which emit parallel light rays, like the sun), and spotlights.

Camera Components

Camera Components define the player's viewpoint in the game. They determine what the player sees when they play the game. In a first-person game, for example, the camera component might be attached to the player character's head. In a third-person game, the camera might follow behind the player character.

Movement Components

Movement Components control how an Actor moves within the game world. For example, a Character Movement Component would control how a player character can walk, jump, or swim.

Colliders

Colliders are another critical type of component in UE5. They define an Actor's physical presence in the world, allowing it to collide with other Actors, triggering events like sound effects, damage, or simply blocking movement. Remember, these are just the basics. UE5 has an extensive library of components that allow you to build complex and detailed game worlds. Each component has its own set of properties and behaviors, allowing you to customize each Actor in your game to behave exactly how you want. 